

Two multiplicative bases in a group algebra

Disproof of Conjecture 00000002758

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The actual group algebra. Let $G = C_2 = \{1, g\}$, $g^2 = 1$, and let $A = \mathbb{Q}[G]$ with its usual convolution multiplication. Its standard group basis is $B = \{1, g\}$. Define

$$e = \frac{1+g}{2}, \quad B' = \{1, e\}.$$

These are two genuine vector-space bases: for the second one, $g = 2e - 1$, and the coordinates of $a + bg$ in B' are $(a - b, 2b)$. The inverse coordinate map is $(s, t) \mapsto s + te$.

Product closure and nonisomorphism. The group basis is closed under multiplication since $g^2 = 1$. In the second basis,

$$e^2 = \frac{1+2g+g^2}{4} = \frac{1+g}{2} = e, \quad 1e = e1 = e.$$

Thus B' is also closed under products, exactly as required by the problem's explicit definition. The coefficient of g in e is $1/2$, whereas it is zero in 1 , so $e \neq 1$. Every element of B has square 1. A multiplication-preserving bijection $B \rightarrow B'$ would map the identity to the identity and hence would force every element of B' to have square 1. This is impossible for e , whose square equals $e \neq 1$. Therefore the two multiplicative bases are not isomorphic as multiplicative structures. In particular no algebra automorphism of A maps B onto B' : if $\phi(b) = e$ for $b \in B$, then $e^2 = \phi(b^2) = \phi(1) = 1$, a contradiction.

The subgroup hypothesis. The group C_2 has exactly two subgroups, $\{1\}$ and C_2 . Indeed any subgroup containing g is the full group, and otherwise contains only the identity. The two subgroups have orders one and two, so they are not isomorphic. Thus there are no distinct isomorphic subgroup pairs, precisely the stated uniqueness hypothesis. Nevertheless the two multiplicative bases above are nonisomorphic, which disproves the conjecture's uniqueness assertion.

Scope of isomorphism. The isomorphisms here preserve the basis multiplication, as required for a meaningful assertion of uniqueness of multiplicative bases. All vector-space bases of the same size admit linear bijections, so forgetting multiplication would make the proposed uniqueness condition vacuous. The source does not require basis elements to be units; imposing that extra requirement would exclude the idempotent basis and would be a different claim.

Formal certificate. Lean uses the actual monoid algebra of the multiplicative group of $\mathbb{Z}/2\mathbb{Z}$, with its genuine convolution product. Both bases are constructed from explicit invertible coordinate maps. Product closure, idempotence and the impossibility of an algebra automorphism between the bases are certified, as is the absence of distinct isomorphic subgroups. Final axiom audits use only standard Lean and Mathlib axioms.